

RYAN SMITH

FULL-STACK SOFTWARE ENGINEER

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Summary

I am a curious and naturally visionary puzzle-solver with a background in theatre and music performance, seeking new opportunities as a full-stack software engineer, especially in the EdTech space. I am fond of working in stacks that include TypeScript and React, love strongly typed, compiled backend languages, and enjoy both functional and object oriented programming. Claude Code and I are a great team.

Experience

Software Engineer

2022 to 2026

Pedago (Quantic School of Business and Technology) • Remote

- Developed full-stack features, for both students and admins, with Rails and Python backend and varied frontend, fixed UX bugs, implemented responsive designs from Sketch and Figma mockups, deployed behind feature flags, reviewed coworkers' pull requests, wrote unit tests and Storybook stories
- Worked with Student Experience team to develop AI experiences and convert legacy AngularJS codebase to React/Redux
- Quickly adapted to new AI tools; used Cursor Agents and Claude Code and completed several large features with quick turnaround; performed independent research to develop original Prompt Engineering techniques; performed self-led data analysis; developed new Claude skills to share with team
- Developed configurations to drive customized UX tailored to a wide range of student/user states
- Implemented DatoCMS-integrated landing pages and many other Gatsby marketing pages on quantic.edu as part of a full rebrand
- Tested our iOS and Android apps in emulators and on-device

Software Developer II

2016 to 2021

Everspring • Chicago, IL

- Developed educational and course authoring software for university online courses, focusing on user experience, scalability, and stakeholder needs
- Canvas LMS and LTI: developed front-end components and single-page applications; architected and implemented back-ends for several applications
- Assisted in user experience design
- Trained new team members

Volunteer / Worship Leader / Musician / Overseas Missionary / Web Developer

2006 to Present

Various Churches • Chicago, IL

- Participated as part of a rotation of worship leaders / support musicians; selected and learned a set of songs for Sunday services; coordinated bandmate schedules and led rehearsals; played piano; led congregation in prayer and singing at church services; led French worship music for children during four mission trips in France; created and/or maintained several church websites; ran audio/visual station during services

Barista / Shift Supervisor

2006 to 2016

Starbucks • Chicago, IL

- Twice named Partner of the Quarter, Winter 2008/9 and Fall 2015; led shifts and supervised the work of fellow employees in high-speed work environment; coached fellow employees toward more efficient customer service; trained new employees; performed cash handling and banking duties; oversaw shift and preparations leading to a rare 100% score in a quality assurance audit

Education

Full Stack Web Development Immersive

2016

Dev Bootcamp

BA Theatre

2006

Northwestern University

Skills

FULL STACK WEB DEVELOPMENT

HTML/CSS/SASS

TypeScript

Ruby on Rails

SQL

PHP

Git/GitHub

RESTful API Integration

Interm. Photoshop

Agile Development

Scala

Spring

NodeJS/NestJS

React/Redux

Angular

Reactive/RxJs

JQuery

Some Rust

Claude Code

Canvas LMS and LTI

AWS: EC2 and S3

Custom AI Agents

Unit Tests

NextJS

Responsive Design

OTHER SKILLS

Piano and Singing

Music Composition

Conversational French and German

Licensed Driver

Projects

AI Advisor / AI Tutor

Python back-end / React front-end. A text and voice chatbot for students to interact with an academic advisor and discuss curriculum and current lesson frame content.

- Implemented chat UI and controls for voice-mode.
- Implemented streaming response tokens over WebSocket connection using reactive programming techniques (Rx).

Student Experience 2.0

Rails back-end / React front-end. Part of a complete overhaul of the in-app student experience, including Admission Application updates, a new Student Dashboard and Sidebar Box system, a complete rebuild of the Student Network, a wide update of app styling, and development of a Slack-like Student Messaging widget.

- Worked with team but completed many features solo, often in discussion with C-suite stakeholders.
- Included conversion from legacy AngularJS frontend to React and TypeScript.

WorldThreads — worldthreads.app

Independent project. Rust back-end / React front-end. Built with Claude. AI chat app where users may create fictional worlds and the characters that inhabit them, chat with characters one-on-one or in groups, generate illustrations and videos, generate novel chapters based on conversations, and chat with a Backstage guide. Intended to feel like an "immersive, living storybook".

- Incorporates novel Prompt Engineering techniques emerging from original research: namely, a math-English hybrid notation that defines the chat completion boundaries (based on first principles, fictional world truth, character truth, and present chat moment) using minimal tokens. Feels to me like a promising finding in prompt compression.
- Bring Your Own OpenAI API Key (BYOK); Tauri desktop application

Untitled Player-Piano AI Ecosystem (In Progress)

Independent. Working with the engineers who built a top-tier automated player-piano system installed in grand pianos, including my own. Currently training an AI model to transcribe and play back, live, from any piano audio. Also training on masterworks for eventual AI composition. Built with Claude.

- The ultimate vision is an all-in-one app for recording, composing, sharing, collaborating, and leveraging AI to compose in certain composers' styles and assist in prompt-driven composition.

CourseBuilder

Java Spring back-end / React front-end. A Learning tool (LTI) that operated closely with the Canvas LMS in order to allow academic program heads to quickly bulk create LMS courses with starter content and manage those courses thereafter.

- I was involved early on with the back-end Spring development, particularly in the launch routine for the LTI and the initial route endpoints.
- Afterward, I was the chief developer on the React front-end, which included a custom virtualized, keyboard-navigable data table and many complex forms with reusable form components + tests.

Atlas Publisher

PHP back-end / Angular front-end. Expanded and upgraded the in-house marketing site authoring experience to feel more like a CMS with authoring forms, users, asset management, and version control.

- The experience allowed non-techie users at the universities to manage their own sites without the aid of developers at Everspring.
- The marquee feature was that users could stage new versions of their site and then promote individual pages to the live site once they were approved.

Branching Scenario Creator

Scala/Play back-end / React front-end. Tool that allowed Instructional Designers to visualize and author a choose-your-own-adventure lesson scenario that branched down different narrative paths based on student decisions.

- The main feature was a tree visualization of the branching content flow.
- Made the authoring experience much easier and led to higher usage of the branching scenario in courses.

Skills and Values Portfolio (SVP)

Scala/Play back-end / React front-end. Learning tool that integrated with course and student data to display student progress through a program of courses, each of which were associated with custom-defined skills and values.

- Students could request feedback for co-curricular activities, see own progress, receive completion badges, and submit periodic reflection essays.
- Program heads and professors could browse student profiles and see summaries of student activity and engagement.

Virtual Community

Scala/Play back-end / React front-end. Learning tool that allowed students to explore a virtual city community by visiting informational pages via a rich map with data overlays.